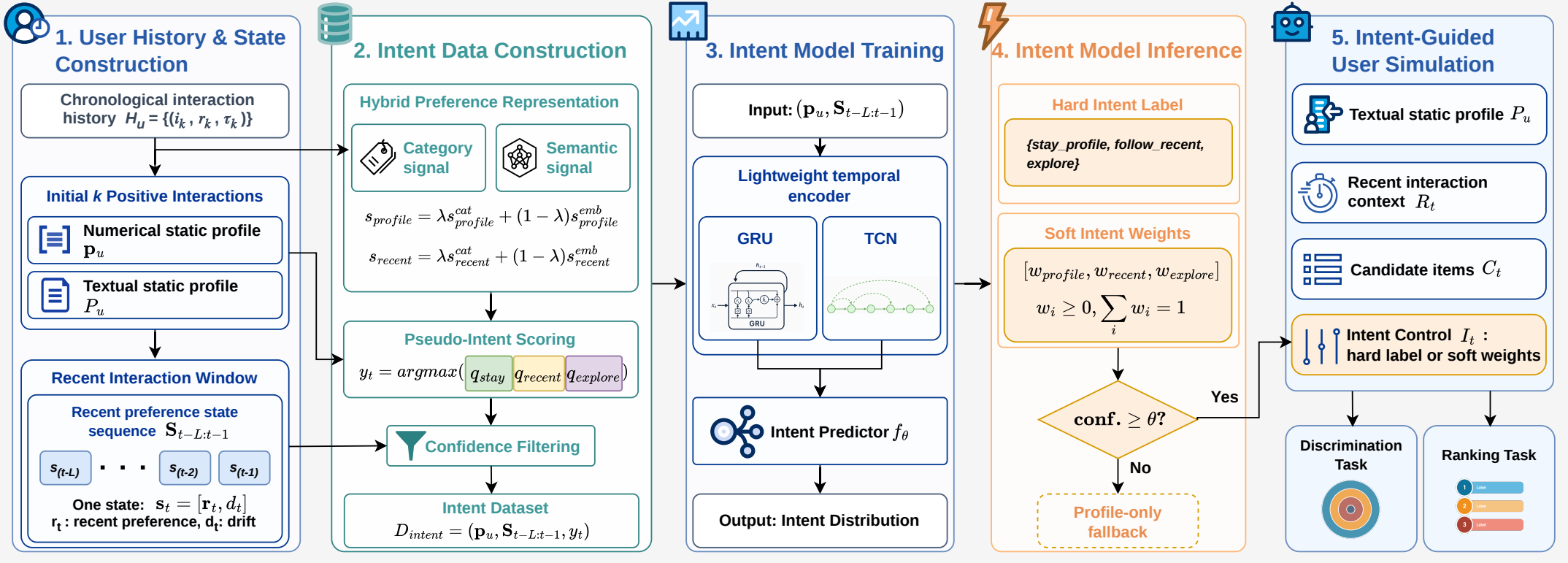

1. User History & State Construction
Chronological interaction history $H_u = \{(i_k, r_k, \tau_k)\}$
Initial k Positive Interactions
Numerical static profile $\mathbf{p}_u$
Textual static profile P_u
Recent Interaction Window
Recent preference state sequence $\mathbf{S}_{t-L:t-1}$
$s_{(t-L)}$ $s_{(t-2)}$ $s_{(t-1)}$
One state: $\mathbf{s}_t = [\mathbf{r}_t, d_t]$
$\mathbf{r}_t$: recent preference, $\mathbf{d}_t$: drift
2. Intent Data Construction
Hybrid Preference Representation
Category signal
Semantic signal
$s_{profile} = \lambda s^{cat}_{profile} + (1-\lambda) s^{emb}_{profile}$
$s_{recent} = \lambda s^{cat}_{recent} + (1-\lambda) s^{emb}_{recent}$
Pseudo-Intent Scoring
$y_t = argmax(q_{stay}\ q_{recent}\ q_{explore})$
Confidence Filtering
Intent Dataset
$D_{intent} = (\mathbf{p}_u, \mathbf{S}_{t-L:t-1}, y_t)$
3. Intent Model Training
Input: $(\mathbf{p}_u, \mathbf{S}_{t-L:t-1})$
Lightweight temporal encoder
GRU
TCN
Intent Predictor f_θ
Output: Intent Distribution
4. Intent Model Inference
Hard Intent Label
{stay_profile, follow_recent, explore}
Soft Intent Weights
$[w_{profile}, w_{recent}, w_{explore}]$
$w_i \geq 0, \sum_i w_i = 1$
conf. $\geq \theta$?
Yes
No
Profile-only fallback
5. Intent-Guided User Simulation
Textual static profile P_u
Recent interaction context R_t
Candidate items C_t
Intent Control I_t : hard label or soft weights
Discrimination Task
Ranking Task